Mathematical Statistics

SNU Statistics graduate entrance exam problems

87 problems arranged by topic, including 40 past exam problems (2020-10 to 2026-04 exams, restated)

Solutions are in a separate booklet.

The exam paper is given in both Korean and English; these problems follow the English version. Past exam problems are tagged [Past exam year-month #number].

Notation: Exp(θ) has mean θ; Gamma(α, β) has shape α and scale β; $\chi^2_\alpha(r)$, $t_\alpha(r)$, $F_\alpha(r_1, r_2)$, z_α are upper-α points. A sequence Y_n is $O_p(1)$ if for every $\epsilon > 0$ there are M, N with $P(|Y_n| > M) < \epsilon$ for all $n \geq N$; $Y_n = o_p(1)$ means $Y_n \xrightarrow{p} 0$; $Y_n = O_p(a_n)$ means $Y_n/a_n = O_p(1)$.

Contents

A. Analysis and linear algebra **2**

Problem 1. Analysis . . . 2
Problem 2. Analysis: limits and series . . . 3
Problem 3. Mean value theorem [2025-10 #3] . . . 4
Problem 4. Uniform convergence [2026-04 #3] . . . 5
Problem 5. Compactness and completeness [2024-10 #3] . . . 6
Problem 6. Linear algebra . . . 7

B. Distributions and transformations **8**

Problem 7. Joint, marginal and conditional distributions . . . 8
Problem 8. Transformations . . . 9
Problem 9. Uniform distribution on a ball [2026-04 #2] . . . 10
Problem 10. Conditional expectation and random sums . . . 11
Problem 11. Poisson process [2022-04 #3] . . . 12
Problem 12. Minimum of a random number of uniforms [2023-04 #3] . . . 13
Problem 13. Moment generating functions and cumulants . . . 14
Problem 14. Uniform order statistics . . . 15
Problem 15. Logistic order statistics [2022-10 #1] . . . 16
Problem 16. Multivariate normal . . . 17
Problem 17. Bivariate normal . . . 18
Problem 18. Multivariate normal with AR(1) covariance [2025-10 #2] . . . 19
Problem 19. Normal samples and prediction . . . 20
Problem 20. Chi-square, t and F . . . 21
Problem 21. Quadratic forms . . . 22
Problem 22. Independence of quadratic forms [2022-10 #5] . . . 23
Problem 23. Distributions of quadratic forms [2022-10 #6] . . . 24

Problem 24. Expectation of a quadratic form [2023-04 #1] . . . 25
Problem 25. Distribution of an F statistic [2024-05 #4] . . . 26

C. Convergence and limit theorems **27**
Problem 26. Stochastic order and limits . . . 27
Problem 27. Tightness [2024-10 #2] . . . 28
Problem 28. O_p and o_p algebra [2025-04 #1] . . . 29
Problem 29. Modes of convergence . . . 30
Problem 30. Probability inequalities . . . 31
Problem 31. Correlation and convergence in distribution [2022-10 #2] . . . 32
Problem 32. Central limit theorem in practice . . . 33
Problem 33. Delta method and variance stabilization . . . 34
Problem 34. Asymptotics of the sample standard deviation [2024-05 #1] . . . 35
Problem 35. Fast convergence of S_n [2024-05 #2] . . . 36
Problem 36. Extremes . . . 37
Problem 37. Asymptotic independence of extremes [2021-10 #5] . . . 38
Problem 38. Limit of an order-statistic functional [2020-10 #1] . . . 39

D. Point estimation **40**
Problem 39. MLE, MME and efficiency . . . 40
Problem 40. Multinomial reparametrization . . . 41
Problem 41. Multinomial reparametrization [2026-04 #1] . . . 42
Problem 42. Beta model: MME and MLE [2025-04 #2] . . . 43
Problem 43. MLE and MME compared by MSE [2025-10 #1] . . . 44
Problem 44. MME and MLE for a uniform model [2023-10 #3] . . . 45
Problem 45. Cauchy location . . . 46
Problem 46. A non-regular scale model . . . 47
Problem 47. Two-parameter exponential . . . 48
Problem 48. Normal with equal mean and variance . . . 49
Problem 49. Two-parameter uniform . . . 50
Problem 50. Minimal sufficiency [2024-05 #3] . . . 51
Problem 51. Sufficiency, completeness, UMVUE . . . 52
Problem 52. Basu's theorem . . . 53
Problem 53. Rao–Blackwellization . . . 54
Problem 54. Cramér–Rao bound . . . 55
Problem 55. UMVUE and the Cramér–Rao bound [2020-10 #3] . . . 56
Problem 56. UMVUE of a normal probability [2023-04 #4] . . . 57
Problem 57. Estimating a Poisson zero probability [2023-10 #1] . . . 58
Problem 58. UMVUE for a uniform region . . . 59
Problem 59. UMVUE for a triangle [2021-04 #5] . . . 60
Problem 60. UMVUE for a disk [2021-10 #4] . . . 61
Problem 61. UMVUE of a ratio of uniform scales [2022-04 #1] . . . 62
Problem 62. Kullback–Leibler divergence . . . 63

Problem 63. Censored data . . . 64
Problem 64. MLE on a restricted parameter space [2021-04 #3] . . . 65

E. Hypothesis testing **66**
Problem 65. Most powerful and UMP tests . . . 66
Problem 66. Power and sample size . . . 67
Problem 67. Tests for a normal variance . . . 68
Problem 68. Likelihood ratio test . . . 69
Problem 69. LRT for a uniform scale . . . 70
Problem 70. Non-uniqueness of UMP tests . . . 71
Problem 71. UMP test for a uniform location . . . 72
Problem 72. Non-uniqueness of MP tests [2024-10 #1] . . . 73
Problem 73. UMP test for a uniform scale [2021-04 #4] . . . 74
Problem 74. UMP test for a Pareto scale [2022-10 #4] . . . 75
Problem 75. UMP test for a logistic location [2022-04 #2] . . . 76
Problem 76. UMP test for a Weibull model [2023-04 #5] . . . 77
Problem 77. Reversing a UMP test [2023-10 #2] . . . 78
Problem 78. Wald, score and LRT . . . 79
Problem 79. LRT on a restricted parameter space . . . 80
Problem 80. LRT on a restricted parameter space [2020-10 #2] . . . 81
Problem 81. LRT for a two-parameter exponential [2021-10 #3] . . . 82

F. Interval estimation and Bayesian inference **83**
Problem 82. Pivots and confidence intervals . . . 83
Problem 83. Bayes and minimax . . . 84
Problem 84. Bayesian prediction and risk . . . 85

G. Nonparametric methods **86**
Problem 85. Signed rank test . . . 86
Problem 86. Signed rank statistic [2025-04 #3] . . . 87
Problem 87. Wilcoxon rank sum . . . 88

A. Analysis and linear algebra

Problem 1

Analysis

(a) Using the mean value theorem, show that $\dfrac{x}{1+x} < \log(1+x) < x$ for all $x > 0$.

(b) Let $f_n(x) = nxe^{-nx}$ for $x \geq 0$. Find the pointwise limit f. Is the convergence uniform on $[0, \infty)$? On $[a, \infty)$ for a fixed $a > 0$?

(c) Show that $\int_0^1 f_n(x)\,dx \to \int_0^1 f(x)\,dx$, even though the convergence in (b) is not uniform on $[0, 1]$.

(d) Let $K \subset \mathbb{R}^k$ be compact and $g : K \to \mathbb{R}$ continuous. Prove that g is uniformly continuous on K.

Problem 2

Analysis: limits and series

(a) Using Problem 1(a), prove $\lim_{n\to\infty}(1 + x/n)^n = e^x$ for $x > 0$.

(b) Show that $\sum_{k=0}^{\infty} x^k/k!$ converges uniformly on $[-a, a]$ for every $a > 0$.

(c) Justify differentiating this series term by term, and conclude that its sum $E(x)$ satisfies $E' = E$.

(d) Show that a function with $f'' \geq 0$ on an interval is convex, using the mean value theorem.

Problem 3 **[Past exam 2025-10 #3]**

Mean value theorem

(a) State and prove the mean value theorem.

(b) Give a function satisfying its hypotheses and explain.

(c) Give a continuous function for which the conclusion fails, and explain why.

Problem 4

[Past exam 2026-04 #3]

Uniform convergence

$f_n(x) = nx(1-x)^n$ on $[0,1]$.

(a) Find the pointwise limit.

(b) Is the convergence uniform?

Problem 5 **[Past exam 2024-10 #3]**

Compactness and completeness

Show that a compact subset of a metric space is complete.

Problem 6

Linear algebra

(a) Show that a symmetric idempotent matrix has eigenvalues 0 or 1 and that its rank equals its trace.

(b) For X of size $n \times p$ with rank p, show that $H = X(X'X)^{-1}X'$ and $I - H$ are symmetric idempotent with traces p and $n - p$, and that $(I - H)X = 0$.

(c) Prove $(I + \mathbf{a}\mathbf{b}')^{-1} = I - \dfrac{\mathbf{a}\mathbf{b}'}{1 + \mathbf{b}'\mathbf{a}}$ when $1 + \mathbf{b}'\mathbf{a} \neq 0$, and $\det(I + \mathbf{a}\mathbf{b}') = 1 + \mathbf{b}'\mathbf{a}$.

(d) Use (c) to invert $\Sigma = (1 - \rho)I_k + \rho\mathbf{1}\mathbf{1}'$, and find for which ρ it is positive definite.

B. Distributions and transformations

Problem 7

Joint, marginal and conditional distributions

(X, Y) is uniformly distributed on $D = \{(x, y) : 0 < x < 1,\ 0 < y < x^2\}$.

(a) Find the joint pdf and the marginal pdfs of X and Y.

(b) Find $E(Y \mid X = x)$ and $\mathrm{Var}(Y \mid X = x)$, and use them to compute EY and $\mathrm{Var}\, Y$.

(c) Compute $\mathrm{Cov}(X, Y)$. Are X and Y independent?

(d) Compute $P(Y > X/2)$.

Problem 8

Transformations

Let $X, Y \overset{\text{iid}}{\sim} \text{Exp}(1)$.

(a) Find the joint pdf of $U = X/(X+Y)$ and $V = X+Y$. Show that they are independent and identify their distributions.

(b) Using (a), find $E(X \mid X+Y=v)$ and $E\big(X^2 \mid X+Y=v\big)$.

(c) Find the pdf of $W = X - Y$.

(d) Let $m = \min(X,Y)$ and $R = \max(X,Y) - \min(X,Y)$. Show that m and R are independent and find their distributions.

Problem 9

[Past exam 2026-04 #2]

Uniform distribution on a ball

(X_1, X_2, X_3) is uniform on the unit ball.

(a) Find the joint pdf of (X_1, X_2).

(b) Find the marginal pdf of X_1.

(c) Find the pdf of X_1 given $X_2 = 0$.

(d) Find $\mathrm{Cov}(\mathbf{X})$.

Problem 10

Conditional expectation and random sums

Let $N \sim \text{Poisson}(\lambda)$ be independent of $X_1, X_2, \ldots$, which are i.i.d. with mean μ, variance σ^2 and mgf M_X. Let $S = \sum_{i=1}^{N} X_i$ ($S = 0$ if $N = 0$).

(a) Find ES and $\text{Var}\, S$.

(b) Show that $M_S(t) = \exp\{\lambda(M_X(t) - 1)\}$.

(c) If $X_i \sim \text{Bernoulli}(p)$, identify the distribution of S.

(d) Show that $E(X_1 \mid S, N = n) = S/n$ for $n \geq 1$.

Problem 11

[Past exam 2022-04 #3]

Poisson process

Buses arrive at a stop as a Poisson process of rate 1 per hour, and each stays 10 minutes. Given that two buses arrived between noon and 1 pm, find the probability that both were at the stop at the same time.

Problem 12

[Past exam 2023-04 #3]

Minimum of a random number of uniforms

$U_1, U_2, \ldots \overset{\text{iid}}{\sim} U(0,1)$, independent of X with $P(X = x) = \frac{c}{x!}$, $x = 1, 2, \ldots$, $c = 1/(e-1)$. For $Z = \min\{U_1, \ldots, U_X\}$, compute $P(Z \le z)$, $0 < z < 1$.

Problem 13

Moment generating functions and cumulants

(a) Identify the distribution with mgf $(0.3 + 0.7e^t)^5$.

(b) $X_1 \sim \text{Poisson}(2)$ and $X_2 \sim \text{Poisson}(1)$ are independent and $Y = X_1 + 2X_2$. Find the mgf, mean and variance of Y. Is Y Poisson?

(c) Show that the r-th cumulant of $\text{Gamma}(\alpha, \beta)$ is $\alpha\beta^r(r-1)!$, and find its skewness.

(d) A random variable satisfies $EX^k = (k+1)!$ for $k = 0, 1, 2, \ldots$. Identify its distribution.

Problem 14

Uniform order statistics

Let $U_{(1)} < \cdots < U_{(n)}$ be the order statistics of $n \geq 2$ i.i.d. $U(0,1)$ variables.

(a) Find the joint pdf of $(U_{(1)}, U_{(n)})$.

(b) Show that the range $R = U_{(n)} - U_{(1)}$ has the Beta$(n-1, 2)$ distribution, and find ER.

(c) Using (b), show that $\mathrm{Cov}(U_{(1)}, U_{(n)}) = \dfrac{1}{(n+1)^2(n+2)}$.

(d) Find $P(U_{(1)} < \frac{1}{2} < U_{(n)})$.

Problem 15

[Past exam 2022-10 #1]

Logistic order statistics

$X_1, \ldots, X_n$ have the logistic pdf $e^x/(1+e^x)^2$. Find the pdf of $\log \dfrac{1+e^{-X_{(1)}}}{1+e^{-X_{(n)}}}$.

Problem 16

Multivariate normal

Let $\mathbf{X} = (X_1, X_2, X_3)' \sim N_3(\mathbf{0}, \Sigma)$ with $\Sigma = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{pmatrix}$.

(a) Find the distribution of $X_1 - X_2 + X_3$.

(b) Find the constant c for which $X_2 - c(X_1 + X_3)$ and $X_1 + X_3$ are independent, and justify independence.

(c) Find the conditional distribution of X_2 given $(X_1, X_3) = (x_1, x_3)$.

(d) Find the mgf $M(t) = E\exp(tX_1X_3)$ and state where it is finite.

(e) What is the distribution of $\mathbf{X}'\Sigma^{-1}\mathbf{X}$? Justify.

Problem 17

Bivariate normal

(X, Y) is bivariate normal with means 0, variances 1 and correlation $\rho \in (-1, 1)$.

(a) Show that $P(X > 0, Y > 0) = \dfrac{1}{4} + \dfrac{\arcsin \rho}{2\pi}$.

(b) Show that $E \max(X, Y) = \sqrt{(1 - \rho)/\pi}$.

(c) Find $E(Y^2 \mid X = x)$ and $\mathrm{Cov}(X^2, Y^2)$.

Problem 18

[Past exam 2025-10 #2]

Multivariate normal with AR(1) covariance

$(X_1, X_2, X_3)' \sim N_3(\mathbf{0}, \Sigma)$ with $\Sigma = \begin{pmatrix} 1 & \rho & \rho^2 \\ \rho & 1 & \rho \\ \rho^2 & \rho & 1 \end{pmatrix}$, $|\rho| < 1$.

(a) Find the joint mgf.

(b) Find $\mathrm{Var}(X_1 - 2X_2 + X_3)$.

(c) Are $X_1 - X_2$ and $X_2 - X_3$ independent?

Problem 19

Normal samples and prediction

Let $X_1, \ldots, X_{n+1} \overset{\text{iid}}{\sim} N(\mu, \sigma^2)$, and let $\bar{X}$ and S^2 be computed from the first n.

(a) Show that $\dfrac{X_{n+1} - \bar{X}}{S\sqrt{1 + 1/n}} \sim t(n-1)$, and give a 95% prediction interval for X_{n+1}.

(b) Show that $ES = \sigma\sqrt{\dfrac{2}{n-1}}\dfrac{\Gamma(n/2)}{\Gamma((n-1)/2)}$.

(c) Find c such that $c\sum_{i=1}^{n}(X_i - \bar{X})^2$ minimizes $E\big(c\sum(X_i - \bar{X})^2 - \sigma^2\big)^2$.

Problem 20

Chi-square, t and F

(a) If $T \sim t(r)$, show $T^2 \sim F(1, r)$.

(b) If $X \sim F(r_1, r_2)$, show $\dfrac{r_1 X}{r_2 + r_1 X} \sim \text{Beta}(r_1/2, r_2/2)$.

(c) $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \text{Exp}(\theta)$ and $Y_1, \ldots, Y_m \overset{\text{iid}}{\sim} \text{Exp}(\theta)$ independently. Show $\bar{X}/\bar{Y} \sim F(2n, 2m)$.

(d) Show that $EX = r_2/(r_2 - 2)$ for $X \sim F(r_1, r_2)$, $r_2 > 2$.

Problem 21

Quadratic forms

Let $\mathbf{Z} \sim N_n(\mathbf{0}, I)$.

(a) For $n = 3$, show that $Q = \frac{1}{2}(Z_1 - Z_2)^2 + \frac{1}{6}(Z_1 + Z_2 - 2Z_3)^2 \sim \chi^2(2)$ and that $Q = \sum_{i=1}^{3}(Z_i - \bar{Z})^2$.

(b) For symmetric A, show $E(\mathbf{Z}'A\mathbf{Z}) = \operatorname{tr} A$ and $\operatorname{Var}(\mathbf{Z}'A\mathbf{Z}) = 2\operatorname{tr}(A^2)$.

(c) Let $J = \frac{1}{n}\mathbf{1}\mathbf{1}'$. Show that $\mathbf{Z}'J\mathbf{Z}$ and $\mathbf{Z}'(I - J)\mathbf{Z}$ are independent, and identify their distributions.

Problem 22

[Past exam 2022-10 #5]

Independence of quadratic forms

$\mathbf{y} \sim N_n(\mu, V)$ with V positive definite, and A, B symmetric idempotent. Show that (1) $\mathbf{y}'A\mathbf{y}$ and $B\mathbf{y}$ are independent iff $BVA = 0$;

(2) $\mathbf{y}'A\mathbf{y}$ and $\mathbf{y}'B\mathbf{y}$ are independent iff $AVB = 0$.

Problem 23

[Past exam 2022-10 #6]

Distributions of quadratic forms

$\mathbf{y} \sim N_n(\mathbf{0}, I)$ and X is $n \times k$ of rank $k < n$, $H = X(X'X)^{-1}X'$. Find the distributions of $\mathbf{y}'H\mathbf{y}$, $\mathbf{y}'(I-H)\mathbf{y}$ and $T = \mathbf{y}'H\mathbf{y}/\mathbf{y}'(I-H)\mathbf{y}$.

Problem 24

[Past exam 2023-04 #1]

Expectation of a quadratic form

For a random vector $\mathbf{y}$ with mean μ and covariance Σ and symmetric A, show $E(\mathbf{y}'A\mathbf{y}) = \operatorname{tr}(A\Sigma) + \mu'A\mu$.

Problem 25

[Past exam 2024-05 #4]

Distribution of an F statistic

$\mathbf{y} \sim N(X\beta, \sigma^2 I_n)$, with X of size $n \times (p+1)$ and full column rank, and H the hat matrix. Find the distribution of $F = \mathbf{y}'H\mathbf{y}/\mathbf{y}'(I-H)\mathbf{y}$.

C. Convergence and limit theorems

Problem 26

Stochastic order and limits

(a) Show that if $X_n \xrightarrow{d} X$, then $X_n = O_p(1)$.

(b) Show that if $X_n = O_p(1)$ and $Y_n = o_p(1)$, then $X_n Y_n = o_p(1)$.

(c) Let $X_1, \ldots, X_n \overset{\text{iid}}{\sim} U(0, \theta)$. Show that $n(\theta - X_{(n)}) \xrightarrow{d} \text{Exp}(\theta)$. Conclude that $X_{(n)} - \theta = O_p(n^{-1})$ and $\sqrt{n}(X_{(n)} - \theta) \xrightarrow{p} 0$.

(d) Let $X_1, \ldots, X_n$ be i.i.d. with mean μ and variance $\sigma^2 < \infty$. Without quoting the delta method, show $\sqrt{n}(\bar{X}_n^2 - \mu^2) \xrightarrow{d} N(0, 4\mu^2\sigma^2)$, using the decomposition $\bar{X}_n^2 - \mu^2 = 2\mu(\bar{X}_n - \mu) + (\bar{X}_n - \mu)^2$.

Problem 27 [Past exam 2024-10 #2]

Tightness

If $X_n \xrightarrow{d} Z$, where all cdfs are continuous, show $X_n = O_p(1)$.

Problem 28

[Past exam 2025-04 #1]

O_p and o_p algebra

Let $X_n, Y_n = O_p(1)$ and $Z_n = o_p(1)$. Prove (1) $X_n + Y_n = O_p(1)$;

(2) $X_n Y_n = O_p(1)$;

(3) $Z_n X_n = o_p(1)$;

(4) $Z_n = O_p(1)$.

Problem 29

Modes of convergence

(a) Let $P(X_n = n) = 1/n$, $P(X_n = 0) = 1 - 1/n$. Show $X_n \xrightarrow{p} 0$ but $EX_n \not\to 0$.

(b) Let $Z \sim N(0,1)$ and $X_n = (-1)^n Z$. Show $X_n \xrightarrow{d} Z$ but X_n does not converge in probability to Z.

(c) Show that if $X_n \xrightarrow{d} c$ for a constant c, then $X_n \xrightarrow{p} c$.

(d) Show that if $E(X_n - a)^2 \to 0$, then $X_n \xrightarrow{p} a$.

Problem 30

Probability inequalities

(a) Using Jensen's inequality, show that $(E|X|^r)^{1/r}$ is nondecreasing in $r > 0$.

(b) Let $\hat{p}$ be the proportion of successes in n Bernoulli(p) trials. Using Chebyshev's inequality, find n that guarantees $P(|\hat{p} - p| \geq 0.05) \leq 0.05$ for every p.

(c) (Chernoff bound) For $X \sim \text{Poisson}(\lambda)$, show $P(X \geq 2\lambda) \leq \inf_{t>0} e^{-2\lambda t} E e^{tX}$, and evaluate the bound.

Problem 31

[Past exam 2022-10 #2]

Correlation and convergence in distribution

$\mathrm{Corr}(X_n, Y_n) \to 1$, and $(X_n - EX_n)/\sqrt{\mathrm{Var}\, X_n} \xrightarrow{d} X$. Show that $(Y_n - EY_n)/\sqrt{\mathrm{Var}\, Y_n} \xrightarrow{d} X$.

Problem 32

Central limit theorem in practice

(a) Approximate $P\big(\sum_{i=1}^{48} U_i > 26\big)$ for $U_i \overset{\text{iid}}{\sim} U(0,1)$.

(b) How large must n be so that a 95% interval $\hat{p} \pm 1.96\sqrt{\hat{p}(1-\hat{p})/n}$ has half-width at most 0.03 regardless of p?

(c) Approximate $P(X \geq 120)$ for $X \sim \text{Poisson}(100)$ with a continuity correction.

Problem 33

Delta method and variance stabilization

(a) For a Poisson(θ) sample, find the limit law of $\sqrt{n}(\sqrt{\bar{X}}-\sqrt{\theta})$ and use it to build an asymptotic 95% interval for θ.

(b) For an Exp(θ) sample, find the limit law of $\sqrt{n}(\log \bar{X} - \log \theta)$.

(c) For $\hat{p} = X/n$, $X \sim B(n,p)$, find the limit law of $\sqrt{n}(\arcsin\sqrt{\hat{p}} - \arcsin\sqrt{p})$.

Problem 34

[Past exam 2024-05 #1]

Asymptotics of the sample standard deviation

For a normal sample, use $(n-1)S_n^2/\sigma^2 \sim \chi^2(n-1)$ to show $\sqrt{n-1}(S_n - \sigma) \xrightarrow{d} N(0, \sigma^2/2)$.

Problem 35

[Past exam 2024-05 #2]

Fast convergence of S_n

X_i are i.i.d. with $P(X_i = \sigma) = P(X_i = -\sigma) = \frac{1}{2}$. Show that $S_n - \sigma = o_p\big((n-1)^{-\gamma}\big)$ for every $\gamma < 1$.

Problem 36

Extremes

Let $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \text{Exp}(1)$.

(a) Show that $X_{(n)} - \log n \xrightarrow{d} G$ with $P(G \leq x) = \exp(-e^{-x})$ (Gumbel).

(b) Find the exact distribution of $nX_{(1)}$.

(c) Show that $nX_{(1)}$ and $X_{(n)} - \log n$ are asymptotically independent.

Problem 37

[Past exam 2021-10 #5]

Asymptotic independence of extremes

$X_1, \ldots, X_n \overset{\text{iid}}{\sim} N(0,1)$. Prove or disprove: $n\Phi(\min X_i)$ and $n\{1 - \Phi(\max X_i)\}$ are asymptotically independent.

Problem 38

[Past exam 2020-10 #1]

Limit of an order-statistic functional

$X_1, \ldots, X_n \overset{\text{iid}}{\sim} N(0,1)$ with order statistics $X_{(1)} < \cdots < X_{(n)}$, $X_{(0)} = -\infty$, $X_{(n+1)} = \infty$. Let

$$T_n = \sum_{i=k+1}^{n} i \int_{X_{(n-i)}}^{X_{(n-i+1)}} \frac{\phi(x)}{1-\Phi(x)} dx, \qquad k < n.$$

(a) Find the distribution of T_n.

(b) For $n = 2k$ and $k \to \infty$, find a_k, b_k such that the mgf of $a_k(T_n - b_k)$ converges to that of $N(0,1)$, and prove it.

D. Point estimation

Problem 39

MLE, MME and efficiency

Let $X_1, \ldots, X_n$ be a sample from the Rayleigh density

$$f(x;\theta) = \frac{x}{\theta}\exp\Big(-\frac{x^2}{2\theta}\Big), \qquad x>0,\ \theta>0.$$

(a) Show that $X_1^2 \sim \mathrm{Exp}(2\theta)$. Find the MLE $\hat\theta$, and its mean and variance.

(b) Compute the Fisher information $I(\theta)$ and show that $\hat\theta$ attains the Cramér–Rao lower bound. Is $\hat\theta$ the UMVUE?

(c) Using $EX_1 = \sqrt{\pi\theta/2}$, find the method-of-moments estimator $\tilde\theta$ based on $\bar X$ and its limiting distribution.

(d) Compute $ARE(\tilde\theta, \hat\theta)$.

Problem 40

Multinomial reparametrization

A sample of n individuals is classified into three genotypes with counts

$$(N_1, N_2, N_3) \sim \text{Multi}\big(n;\ \theta^2,\ 2\theta(1-\theta),\ (1-\theta)^2\big), \qquad 0 < \theta < 1.$$

(a) Find the MLE $\hat{\theta}$.

(b) Show that $2N_1 + N_2 \sim B(2n, \theta)$, and find $E\hat{\theta}$ and $\text{Var}\,\hat{\theta}$.

(c) Find the Fisher information per individual and the limiting distribution of $\sqrt{n}(\hat{\theta} - \theta)$. Is $\hat{\theta}$ the UMVUE?

(d) With $n = 200$ and $(N_1, N_2, N_3) = (50, 90, 60)$, test at level 0.05 whether the three-cell model above fits, against a general trinomial, using Pearson's statistic. State the degrees of freedom.

Problem 41

[Past exam 2026-04 #1]

Multinomial reparametrization

$(X_1, X_2, X_3) \sim \text{Multi}(n; p_1, p_2, p_3)$, reparametrized by $\theta_1 = p_1$, $\theta_2 = p_2/(p_2 + p_3)$.

(a) Find the MLE.

(b) Find the Fisher information matrix.

(c) Find the limit law of $(\hat{\theta}_1, \hat{\theta}_2)$.

Problem 42

[Past exam 2025-04 #2]

Beta model: MME and MLE

For a Beta$(\alpha, 1)$ sample:

(1) show $\hat{\alpha}^{\mathrm{MME}} = \bar{X}/(1-\bar{X})$ and find its limit law;

(2) find the MLE and its limit law;

(3) compute the ARE of the MME with respect to the MLE.

Problem 43

[Past exam 2025-10 #1]

MLE and MME compared by MSE

$X_1, \ldots, X_n$ have pdf $2x/\theta^2$ on $(0, \theta)$.

(a) Find the MLE and the MME.

(b) Find their means and variances.

(c) Compare them by MSE.

Problem 44

[Past exam 2023-10 #3]

MME and MLE for a uniform model

$X_1, \ldots, X_n$ are i.i.d. uniform on $(\theta, \theta + |\theta|)$, $\theta \neq 0$. Find the MME and the MLE of θ, and decide whether the MLE is consistent.

Problem 45

Cauchy location

Let $X_1, \ldots, X_n$ be a sample from the Cauchy density $\frac{1}{\pi}\{1 + (x - \theta)^2\}^{-1}$.

(a) Using the characteristic function $Ee^{itX} = e^{i\theta t - |t|}$, show that $\bar{X}$ has the same distribution as X_1. Is $\bar{X}$ consistent?

(b) Find the limit law of $\sqrt{n}(\hat{m}_n - \theta)$, where $\hat{m}_n$ is the sample median.

(c) Compute the Fisher information $I(\theta)$ and the ARE of the median relative to an efficient estimator.

Problem 46

A non-regular scale model

Let $X_1, \ldots, X_n$ be a sample from $U[\theta, 2\theta]$, $\theta > 0$.

(a) Find the MLE of θ and a minimal sufficient statistic.

(b) Find an unbiased estimator of θ that is a multiple of the MLE.

(c) Show that $(X_{(1)}, X_{(n)})$ is not complete.

Problem 47

Two-parameter exponential

Let $X_1, \ldots, X_n$ $(n \geq 2)$ be a sample from $f(x; \mu, \sigma) = \frac{1}{\sigma} e^{-(x-\mu)/\sigma}$, $x \geq \mu$.

(a) Find the MLEs $\hat{\mu}, \hat{\sigma}$.

(b) Show that $n(\hat{\mu} - \mu)/\sigma \sim \text{Exp}(1)$ and $2n\hat{\sigma}/\sigma \sim \chi^2(2n-2)$, independently.

(c) Given that $(X_{(1)}, \sum X_i)$ is complete sufficient, find the UMVUEs of σ and μ.

Problem 48

Normal with equal mean and variance

Let $X_1, \ldots, X_n$ be a sample from $N(\theta, \theta)$, $\theta > 0$.

(a) Show that the MLE is $\hat{\theta} = \frac{1}{2}\big(-1 + \sqrt{1 + 4m_2}\big)$, where $m_2 = \frac{1}{n}\sum X_i^2$.

(b) Compute $I(\theta)$ and the limit law of $\sqrt{n}(\hat{\theta} - \theta)$.

(c) Compare $\hat{\theta}$ with $\bar{X}$ and S^2 by ARE.

Problem 49

Two-parameter uniform

$X_1, \ldots, X_n$ ($n \geq 2$) are i.i.d. $U(\theta_1, \theta_2)$, $\theta_1 < \theta_2$. You may use that $(X_{(1)}, X_{(n)})$ is complete sufficient.

(a) Find $EX_{(1)}$ and $EX_{(n)}$.

(b) Find the UMVUEs of $\theta_2 - \theta_1$, of $(\theta_1 + \theta_2)/2$, and of θ_1.

Problem 50

[Past exam 2024-05 #3]

Minimal sufficiency

For a sample from $U(\theta, \theta+1)$, $n \geq 2$, find a minimal sufficient statistic.

Problem 51

Sufficiency, completeness, UMVUE

Let $X_1, \ldots, X_n$ be a sample from $f(x;\theta) = e^{-(x-\theta)}$, $x > \theta$, $\theta \in \mathbb{R}$.

(a) Show that $X_{(1)}$ is sufficient, and find its distribution.

(b) Show that $X_{(1)}$ is complete.

(c) Find the UMVUEs of θ and of θ^2.

(d) Show that $X_{(1)}$ and $S = \sum_{i=1}^{n}(X_i - X_{(1)})$ are independent, and find the distribution of S.

Problem 52

Basu's theorem

Let $X_1, \ldots, X_n \overset{\text{iid}}{\sim} N(\mu, \sigma^2)$ with σ^2 known.

(a) Use Basu's theorem to show that $\bar{X}$ is independent of S^2 and of $X_{(n)} - X_{(1)}$.

(b) Show that $E\big(X_{(n)} \mid \bar{X}\big) = \bar{X} + c_n\sigma$ for a constant c_n not depending on μ.

(c) Find $E(X_1^2 \mid \bar{X})$.

Problem 53

Rao–Blackwellization

(a) For a Poisson(θ) sample, compute $E\big(\mathbf{1}\{X_1 = 0\} \mid T\big)$, $T = \sum X_i$, and conclude.

(b) For a Bernoulli(θ) sample ($n \geq 2$), find the UMVUE of θ^2 by conditioning $X_1 X_2$ on T.

(c) For $U(0, \theta)$, improve $2X_1$ by conditioning on $X_{(n)}$.

Problem 54

Cramér–Rao bound

Let $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \text{Poisson}(\theta)$ and $\eta = e^{-\theta} = P(X_1 = 0)$.

(a) Find the Cramér–Rao lower bound for unbiased estimators of η.

(b) The UMVUE of η is $(1 - 1/n)^T$, $T = \sum X_i$. Find its variance and show that it exceeds the bound for all $\theta > 0$.

(c) Find the limit law of $\sqrt{n}(e^{-\bar{X}} - e^{-\theta})$ and compare.

Problem 55

[Past exam 2020-10 #3]

UMVUE and the Cramér–Rao bound

For a sample from $N(\mu, \sigma^2)$ with σ^2 known, find the UMVUE of $e^{c\mu}$ ($c \neq 0$ known), and decide whether it attains the Cramér–Rao bound.

Problem 56

[Past exam 2023-04 #4]

UMVUE of a normal probability

$X_1, \ldots, X_n \overset{\text{iid}}{\sim} N(\theta, 1)$, $n > 1$. Show that $u(X_1) = \mathbf{1}\{X_1 \leq c\}$ is unbiased for $\Phi(c - \theta)$, and use it to find the UMVUE of $\Phi(c - \theta)$.

Problem 57

[Past exam 2023-10 #1]

Estimating a Poisson zero probability

For a Poisson(λ) sample, estimate $\theta = e^{-\lambda}$ by $\hat{\theta}^{(1)} = e^{-\bar{X}}$ or $\hat{\theta}^{(2)} = \frac{1}{n}\sum \mathbf{1}\{X_i = 0\}$. Find both limit laws and decide which is more efficient.

Problem 58

UMVUE for a uniform region

$(X_1, Y_1), \ldots, (X_n, Y_n)$ are i.i.d. uniform on the square $\{(x, y) : |x| + |y| < \theta\}$, $\theta > 0$. Let $W_i = |X_i| + |Y_i|$ and $M = \max_i W_i$.

(a) Find the distribution of W_1, and show that M is sufficient.

(b) Show that M is complete.

(c) Find the UMVUEs of θ and of the area $2\theta^2$.

Problem 59

[Past exam 2021-04 #5]

UMVUE for a triangle

(X_i, Y_i), $i \leq n$, are i.i.d. uniform on the triangle $\{x > 0, y > 0, x + y < \theta\}$. Find the UMVUE of θ and its pdf.

Problem 60

[Past exam 2021-10 #4]

UMVUE for a disk

(X_i, Y_i) are i.i.d. uniform in the disk of radius r centred at the origin, and $\theta = r^2$. Find the UMVUE of θ and its pdf.

Problem 61

[Past exam 2022-04 #1]

UMVUE of a ratio of uniform scales

$X_1, \ldots, X_m \overset{\text{iid}}{\sim} U(0, \theta)$ and, independently, $Y_1, \ldots, Y_n \overset{\text{iid}}{\sim} U(0, \eta)$. Find the UMVUE of θ/η.

Problem 62

Kullback–Leibler divergence

(a) Compute $KL(\theta_1, \theta_0) = E_{\theta_0} \log\{f(X;\theta_0)/f(X;\theta_1)\}$ for $N(\theta, 1)$ and for Poisson(θ).

(b) Using $\log t \le t - 1$, show $KL \ge 0$ with equality iff the densities coincide.

(c) Explain why this suggests that the MLE is consistent.

Problem 63

Censored data

Lifetimes $X_i \overset{\text{iid}}{\sim} \text{Exp}(\theta)$ are censored at a fixed time $c > 0$: we observe $Z_i = \min(X_i, c)$ and $\delta_i = \mathbf{1}\{X_i \le c\}$, $i = 1, \dots, n$.

(a) Write the likelihood and show that $(D, T) = (\sum \delta_i, \sum Z_i)$ is sufficient.

(b) Find the MLE of θ (when $D > 0$).

(c) Show that $I_n(\theta) = nP_\theta/\theta^2$ with $P_\theta = 1 - e^{-c/\theta}$, and give the limit law of the MLE. How much information is lost by censoring?

Problem 64

[Past exam 2021-04 #3]

MLE on a restricted parameter space

$X_1, \ldots, X_n$ have pdf $\theta^{-1}e^{-x/\theta}$, $x > 0$, where the parameter space is $\theta \geq 1$. Find the MLE and the limit law of $\sqrt{n}(\hat{\theta} - \theta)$.

E. Hypothesis testing

Problem 65

Most powerful and UMP tests

Let $X_1, \ldots, X_n$ be a sample from the Pareto density $f(x;\theta) = \theta x^{-\theta-1}$, $x > 1$, $\theta > 0$, and let $T = \sum_{i=1}^{n} \log X_i$.

(a) Show that $2\theta T \sim \chi^2(2n)$.

(b) Find the most powerful level-α test of $H_0 : \theta = 1$ vs $H_1 : \theta = \theta_1$ for a fixed $\theta_1 > 1$.

(c) Show that the test in (b) is UMP for $H_0 : \theta \leq 1$ vs $H_1 : \theta > 1$, and express its power function with a chi-square cdf.

(d) For $n = 10$ and $\alpha = 0.05$ ($\chi^2_{0.95}(20) = 10.851$), give the rejection region and the power at $\theta = 2$.

Problem 66

Power and sample size

$X_1, \ldots, X_n \overset{\text{iid}}{\sim} N(\mu, 4^2)$; test $H_0 : \mu \leq 10$ vs $H_1 : \mu > 10$ at $\alpha = 0.05$.

(a) Give the UMP test and its power function.

(b) Find the smallest n with power at least 0.9 at $\mu = 12$ ($z_{0.10} = 1.282$).

(c) With that n, compute the power at $\mu = 11$.

(d) Show that no UMP test exists for $H_0 : \mu = 10$ vs $H_1 : \mu \neq 10$.

Problem 67

Tests for a normal variance

Let $X_1, \ldots, X_n \overset{\text{iid}}{\sim} N(\mu, \sigma^2)$.

(a) With μ known, find the UMP level-α test of $H_0 : \sigma^2 \leq \sigma_0^2$ vs $H_1 : \sigma^2 > \sigma_0^2$, and its power function.

(b) With μ unknown, show that the LRT of $H_0 : \sigma^2 = \sigma_0^2$ vs $H_1 : \sigma^2 \neq \sigma_0^2$ is based on $v = \hat{\sigma}^2/\sigma_0^2$ through $n(v - 1 - \log v)$, and describe the rejection region.

Problem 68

Likelihood ratio test

Let $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \text{Exp}(\theta_1)$ and $Y_1, \ldots, Y_m \overset{\text{iid}}{\sim} \text{Exp}(\theta_2)$ be independent samples. Consider $H_0 : \theta_1 = \theta_2$ vs $H_1 : \theta_1 \neq \theta_2$.

(a) Show that $2 \log \Lambda$, the LRT statistic, is a function of $B = n\bar{X}/(n\bar{X} + m\bar{Y})$, and describe the form of the rejection region.

(b) Find the exact null distribution of B, and of $\bar{X}/\bar{Y}$.

(c) State the limiting null distribution of $2 \log \Lambda$ as $n, m \to \infty$.

Problem 69

LRT for a uniform scale

Let $X_1, \ldots, X_n \overset{\text{iid}}{\sim} U(0, \theta)$; test $H_0 : \theta = \theta_0$ vs $H_1 : \theta \neq \theta_0$.

(a) Find the LRT statistic and the size-α rejection region.

(b) Show that under H_0, $2 \log \Lambda \sim \chi^2(2)$ exactly. Why is this not $\chi^2(1)$?

Problem 70

Non-uniqueness of UMP tests

$X_1, \ldots, X_n$ are i.i.d. $U(0, \theta)$; test $H_0 : \theta \le \theta_0$ vs $H_1 : \theta > \theta_0$ at level α. Show that both

$$\phi_1 = \mathbf{1}\{X_{(n)} > \theta_0(1-\alpha)^{1/n}\} \qquad \text{and} \qquad \phi_2 = \mathbf{1}\{X_{(n)} > \theta_0\} + \alpha\,\mathbf{1}\{X_{(n)} \le \theta_0\}$$

are UMP, and that they have the same power function.

Problem 71

UMP test for a uniform location

$X_1, \ldots, X_n$ are i.i.d. $U(\theta, \theta+1)$. Test $H_0 : \theta = 0$ vs $H_1 : \theta > 0$ at level $\alpha \in (0,1)$.

(a) For a fixed $\theta_1 \in (0,1)$, compute the likelihood ratio $p_{\theta_1}(\mathbf{x})/p_0(\mathbf{x})$ on each part of the sample space.

(b) Show that the test rejecting when $X_{(n)} \geq 1$ or $X_{(1)} \geq c$, with $c = 1 - \alpha^{1/n}$, has size α and is UMP.

(c) Find its power function.

Problem 72

[Past exam 2024-10 #1]

Non-uniqueness of MP tests

For $U(0,\theta)$ and $H_0 : \theta = \theta_0$ vs $H_1 : \theta = \theta_1$ ($\theta_0 < \theta_1$), show that any test with $\phi = 1$ when $x_{(n)} > \theta_0$ and $E_{\theta_0}\phi = \alpha$ is most powerful.

Problem 73

[Past exam 2021-04 #4]

UMP test for a uniform scale

$X_1, \ldots, X_n \overset{\text{iid}}{\sim} U[0, \theta]$. Find a UMP test of $H_0 : \theta = \theta_0$ vs $H_1 : \theta \neq \theta_0$.

Problem 74

[Past exam 2022-10 #4]

UMP test for a Pareto scale

$X_1, \ldots, X_n$ have the Pareto pdf $\theta a^\theta x^{-(\theta+1)}$, $x > a$, with $\theta > 0$ known. Find a level-α UMP test of $H_0 : a = a_0$ vs $H_1 : a \neq a_0$.

Problem 75

[Past exam 2022-04 #2]

UMP test for a logistic location

A single X has pdf $e^{x-\theta}/(1+e^{x-\theta})^2$. Show that the UMP level-α test of $H_0 : \theta \leq 0$ vs $H_1 : \theta > 0$ rejects when $X > C$, and find C.

Problem 76

[Past exam 2023-04 #5]

UMP test for a Weibull model

$X_1, \ldots, X_n$ have the Weibull pdf $\lambda c x^{c-1} e^{-\lambda x^c}$, $x > 0$, c known.

(1) Show that X_i^c is exponential.

(2) Find the UMP level-α test of $H_0 : \lambda \geq \lambda_0$ vs $H_1 : \lambda < \lambda_0$.

Problem 77

[Past exam 2023-10 #2]

Reversing a UMP test

ϕ is the UMP size-α test of $H_0 : f = f_0$ vs $H_1 : f = f_1$, with power $\beta \in (0, 1)$. Find the UMP size-$(1-\beta)$ test of $H_0^* : f = f_1$ vs $H_1^* : f = f_0$ in terms of ϕ.

Problem 78

Wald, score and LRT

$X \sim B(100, p)$ is observed as $x = 60$; test $H_0 : p = 0.5$ vs $H_1 : p \neq 0.5$ at level 0.05.

(a) Compute the LRT, Wald and score statistics.

(b) State the common limiting null distribution and the decisions.

(c) Which of the three needs only H_0, not the MLE? Give the corresponding 95% confidence set for p (the score interval) as an inequality.

Problem 79

LRT on a restricted parameter space

$X_1, \dots, X_n \overset{\text{iid}}{\sim} N(\mu, 1)$ with parameter space $\mu \geq 0$. Test $H_0 : \mu = 0$ vs $H_1 : \mu > 0$.

(a) Find the MLE and show $2 \log \Lambda = n(\bar{X}^+)^2$, with $a^+ = \max(a, 0)$.

(b) Show that under H_0, $2 \log \Lambda$ has the mixture law $\frac{1}{2}\delta_0 + \frac{1}{2}\chi^2(1)$. Find the size-$\alpha$ critical value ($\alpha < \frac{1}{2}$) and compare with the naive $\chi^2(1)$ calibration.

(c) Find the limit law of $\sqrt{n}(\hat{\mu} - \mu)$ for $\mu > 0$ and for $\mu = 0$.

Problem 80

[Past exam 2020-10 #2]

LRT on a restricted parameter space

$X_i \sim N(\mu_i, 1)$, $i = 1, 2$, independent, with parameter space $\Omega = \{\mu_1 \geq 0, \mu_2 \geq 0\}$. Derive the LRT of $H_0 : (\mu_1, \mu_2) = (0, 0)$ vs $H_1 : (\mu_1, \mu_2) \neq (0, 0)$ and the condition determining its rejection region.

Problem 81

[Past exam 2021-10 #3]

LRT for a two-parameter exponential

$X_1, \ldots, X_n$ have pdf $\sigma^{-1} e^{-(x-\mu)/\sigma}$, $x \geq \mu$, with both parameters unknown. Find the LRT of $H_0 : \mu = \mu_0$ vs $H_1 : \mu \neq \mu_0$.

F. Interval estimation and Bayesian inference

Problem 82

Pivots and confidence intervals

(a) For an $\mathrm{Exp}(\theta)$ sample, give an exact $1-\alpha$ confidence interval for θ.

(b) For a $U(0,\theta)$ sample, show that among intervals $[X_{(n)}/b, X_{(n)}/a]$ with $0 < a < b \leq 1$ and coverage $1-\alpha$, the shortest is $[X_{(n)}, X_{(n)}\alpha^{-1/n}]$.

(c) For $N(\mu,\sigma^2)$ with both unknown, give an exact interval for σ.

Problem 83

Bayes and minimax

Let $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \text{Poisson}(\theta)$ with prior $\theta \sim \text{Exp}(\beta)$, i.e. $\text{Gamma}(1, \beta)$. Let $T = \sum X_i$ and $s = (n + 1/\beta)^{-1}$.

(a) Find the posterior distribution of θ.

(b) Find the Bayes estimator under squared error loss, and write it as a weighted average of $\bar{X}$ and the prior mean.

(c) Under the loss $L(\theta, a) = (a - \theta)^2/\theta$, show that the Bayes estimator is $a^* = sT$ and that its posterior expected loss is s for every T.

(d) Show that $\bar{X}$ is minimax for the loss in (c).

Problem 84

Bayesian prediction and risk

(a) Bernoulli(θ) trials with prior Beta(α, β) give t successes in n. Find $P(X_{n+1} = 1 \mid \mathbf{x})$. What is it for a uniform prior and $t = n$?

(b) $X_1, \ldots, X_n \overset{\text{iid}}{\sim} N(\theta, \sigma^2)$, σ^2 known, prior $N(\mu_0, \tau^2)$. Show that the posterior mean has Bayes risk $(n/\sigma^2 + 1/\tau^2)^{-1}$, smaller than the Bayes risk σ^2/n of $\bar{X}$.

(c) In (b), find the frequentist MSE of the posterior mean as a function of θ, and show that it exceeds σ^2/n when $|\theta - \mu_0|$ is large.

G. Nonparametric methods

Problem 85

Signed rank test

Let $X_1, \ldots, X_n$ be a sample from a continuous distribution symmetric about θ, and $W^+ = \sum_i \mathbf{1}\{X_i > 0\} R_i$, where R_i is the rank of $|X_i|$ among $|X_1|, \ldots, |X_n|$. Under $H_0 : \theta = 0$:

(a) Show that $W^+ \overset{d}{=} \sum_{j=1}^{n} jB_j$ with $B_j \overset{\text{iid}}{\sim}$ Bernoulli(1/2). (You may use that the signs and the ranks of $|X_i|$ are independent.)

(b) Find EW^+ and $\operatorname{Var} W^+$, and the exact null distribution of W^+ for $n = 4$.

(c) Show that $W^+ = \sum_{1 \le i \le j \le n} \mathbf{1}\{X_i + X_j > 0\}$.

(d) The ARE of the signed rank test relative to the t test is $12\sigma^2\big(\int f^2\big)^2$, where f is the density of $X_1 - \theta$ and σ^2 its variance. Evaluate it for f uniform on $(-1, 1)$ and for $f(x) = \frac{1}{2}e^{-|x|}$.

Problem 86

[Past exam 2025-04 #3]

Signed rank statistic

For the signed rank statistic $W_n = \sum \operatorname{sgn}(X_i - \theta_0) R(|X_i - \theta_0|)$ under $H_0 : \theta = \theta_0$ (symmetric continuous f):

(1) show $W_n \overset{d}{=} \sum_j j S_j$ with i.i.d. fair signs S_j;

(2) show $W_n / \sqrt{\operatorname{Var} W_n} \xrightarrow{d} N(0,1)$;

(3) show $W_n^+ \overset{d}{=} \sum_j j B_j$ with $B_j \overset{\text{iid}}{\sim} \text{Bernoulli}(\frac{1}{2})$.

Problem 87

Wilcoxon rank sum

Let $X_1, \ldots, X_m$ and $Y_1, \ldots, Y_n$ be independent samples from continuous distributions, $N = m+n$, and $W = \sum_j R(Y_j)$, the sum of the ranks of the Y's in the pooled sample.

(a) Under H_0 (same distribution), show that every rank R has $ER = \frac{N+1}{2}$, $\operatorname{Var} R = \frac{N^2-1}{12}$, and that two distinct ranks have covariance $-\frac{N+1}{12}$.

(b) Deduce $EW = \frac{n(N+1)}{2}$ and $\operatorname{Var} W = \frac{mn(N+1)}{12}$.

(c) For $m = n = 3$, find $P(W \geq 14)$ under H_0.